Genius Olympiad: Research Paper
Project Title: Autonomous Irrigation System
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Abstract

Agriculture plays a critical role in global food systems and improving irrigation efficiency is essential for environmental and economic sustainability. In India, agriculture employs approximately 40% of the population but contributes only about 17% to the national GDP, highlighting the continued reliance on labor-intensive farming practices. This situation is particularly significant in a country where agriculture consumes a large proportion of freshwater resources, making efficient irrigation methods increasingly important.

The purpose of this study was to design and evaluate an affordable autonomous irrigation system that assists farmers by monitoring soil conditions and regulating irrigation more efficiently than conventional methods. The system was developed using soil moisture sensors connected to ESP32 microcontrollers that continuously measure soil moisture levels. Sensor data is transmitted to a central control hub, which processes the readings and activates a water pump through a relay module when soil moisture drops below a predefined threshold. The device also incorporates an OLED display to provide real-time status updates, while a mobile application interface enables remote monitoring and manual control of irrigation.

Results confirmed that the device can reliably monitor soil conditions and automate water delivery with minimal human intervention. Water consumption was reduced by 20%, while plant growth rate increased by 15%. Farmers also reported reduced involvement since crop watering could be done remotely, allowing better use of their time thus improving their lifestyle.

Introduction and Goal

Agriculture remains the backbone of India's economy, employing over 40% of the population but contributing only about 17% to the country's GDP. Despite being one of the largest agricultural producers globally, India's farming sector still relies heavily on manual labor and outdated techniques, unlike countries like the Netherlands, Israel, and the USA, where automation and precision agriculture have revolutionized farming efficiency. During a visit to a

local farm, I witnessed firsthand the difficulties faced by Indian farmers, particularly fruit and vegetable growers who spend hours each day manually irrigating their fields. This daily burden forces many to work over 12 hours under harsh conditions, with little technological support. With agriculture consuming nearly 70% of the world's freshwater resources, improving irrigation efficiency is not just a local challenge but a global necessity. This project seeks to provide a small but meaningful step toward addressing that gap: reducing manual labor for farmers while conserving water in one of the most water-stressed countries in the world.

Literature Review

The concept of autonomous irrigation systems is gaining prominence globally as agriculture shifts toward more efficient, technology-driven solutions. Multiple studies have shown that integrating real-time environmental sensing and IoT can significantly reduce water usage and improve crop yields. Research papers such as "*A Novel Autonomous Irrigation System for Smart Agriculture Using AI and 6G-IoT*" and "*The Effects of AI-Supported Autonomous Irrigation Systems on Water Efficiency and Plant Quality*" validate this approach by highlighting how data-driven irrigation scheduling leads to better water conservation and healthier crops. These sources informed the development of this project by demonstrating the feasibility and effectiveness of automated systems, which influenced both the system design and the parameters chosen for testing. Additionally, technical resources from Arduino.cc, SparkFun, and Adafruit were instrumental in the hands-on development of the project, enabling robust understanding of electronics and embedded programming.

My aim is to build a cost-effective, autonomous system by optimizing water irrigation based on soil moisture levels through IoT

1. Improve quality of life for farmers by reducing hours spent on the field
2. Reduce water wastage by adapting water quantity dispensed according to environmental conditions
3. Improve crop yield by optimizing irrigation
4. Create an easy to install module which can be operated with rudimentary knowledge of technology
5. Conduct a pilot study to compare crop yield, quality of life of farmers and water wastage with and without the module
6. Enable farmers with modern technology

Methods

Timeline

I chose this structured, phased approach to ensure that each stage of the project built logically on the one before it. By starting with problem identification and research before moving into practical work and testing, I was able to balance learning, experimentation, and implementation efficiently. The timeline was designed to give flexibility for trial and error in the middle phases while ensuring that enough time remained at the end for real-world testing, analysis, and reporting. This method allowed for steady progress while minimizing risks of delays or incomplete work.

Development process

The development of the Autonomous Irrigation System was carried out 6 months and involved a structured, hands-on learning and prototyping process. The project began with preliminary research and surveys to understand the real-world problems faced by small-scale farmers, particularly around inefficient water use. This helped in the identification of the aim and problem statement, which focused on creating a smart, affordable system to automate irrigation using real-time soil moisture data.

Following this, a significant phase of skill-building was undertaken. I engaged in guided sessions to understand the basics of electrical components such as resistors, LEDs, sensors, pumps, and relays. I practiced on introductory circuits using breadboards and Tinkercad simulations starting with simple projects like blinking LEDs using digital output, reading sensor data through analog input, and using control structures like if-else statements to automate LED signals based on sensor readings.

Once I was comfortable with component behavior, I moved on to Arduino programming, learning about data types, constants, logic integration, and coding using the Arduino IDE. This was followed by project planning, where I mapped out the full system in a flowchart, breaking the process into input (sensor data), logic (threshold-based decision-making), and output (pump activation). Based on this, I finalized a component list with the help of my mentor and procured necessary hardware, including soil moisture sensors, submersible pumps, relay modules, OLED displays, ESP32 boards, rechargeable batteries, and supporting electronics.

Prototyping began with integrating individual modules, first building a soil moisture sensor circuit to read and display real-time values using LEDs and OLED screens, then developing a motor activation module using relays. These were then combined and refined through several iterations, addressing issues like inconsistent readings, response delays, and motor calibration.

Parallel to this, I worked on the water delivery method, settling on a simple submersible pump setup suited for small-scale irrigation.



Fig: Hardware Prototype

As the circuit stabilized, I focused on code integration, ensuring that sensor data triggered the pump precisely and that feedback was shown on the display. Later, I transitioned to the ESP32 microcontroller, learning to install board libraries, and began exploring Blynk platform integration for app-based monitoring and control. Communication between nodes and hubs using the ESP32 C3 Mini board was tested, with further work planned for future expansion. After working on soil sensing nodes, I programmed and built the autonomous water dispensing system to trigger 3-6V Submersible Mini Water Pump when soil moisture drops to a predefined level. Throughout the process, I built low-fidelity prototypes for enclosure design using cardboard and finally settled on a simple design using a 3D printed box attached to a PVC pipe. The journey concluded with the assembly of a fully functional model, demonstrating an end-to-end automated irrigation solution with real-time sensing, feedback display, and scalability potential.

Description of Implementation

The implementation of this autonomous irrigation system allows farmers to interact with their fields through a user-friendly combination of hardware modules and the Blynk mobile app. Multiple soil moisture sensors placed across different zones of the field continuously collect real-time data, enabling the system to identify which areas require watering.

```
void loop() {
  // Read sensor data (simulate sensor for testing)
  message.ID = SENDER_ID;
  int sensorValue = analogRead(6);  // Use correct pin for your setup
  message.integer = map(sensorValue, 0, 4095, 0, 100);  // Map to percentage

  esp_err_t result = esp_now_send(broadcastAddress, (uint8_t *)&message, sizeof(message));

  Serial.printf("Sender ID %d sent: %d%% moisture\n", message.ID, message.integer);

  // Update OLED
  display.clearDisplay();
  display.setTextSize(1);
  display.setTextColor(WHITE);

  display.setCursor(0, 0);
  display.print("Connection: ");
  display.println(lastSendSuccess ? "OK" : "X");

  display.setCursor(0, 16);
  display.print("Soil Moisture: ");
  display.print(message.integer);
  display.println("%");

  display.display();

  delay(6000);  // Send interval (adjust if necessary)
```

Based on this input, the ESP32-C3 microcontroller activates specific water pumps connected via relay modules to target only the zones that need moisture, ensuring efficient water use.

```
void setup() {
  Serial.begin(115200);
  WiFi.setSleep(false);  // Disable power-saving for better stability
  WiFi.mode(WIFI_AP_STA);  // AP for ESP-NOW, STA for Blynk

  // Relay setup
  pinMode(RELAY1_PIN, OUTPUT);
  pinMode(RELAY2_PIN, OUTPUT);
  pinMode(RELAY3_PIN, OUTPUT);
  digitalWrite(RELAY1_PIN, HIGH);
  digitalWrite(RELAY2_PIN, HIGH);
  digitalWrite(RELAY3_PIN, HIGH);

  // Connect to router (STA mode)
  WiFi.begin(ssid, pass);
  Serial.print("[WiFi] Connecting to router");
  while (WiFi.status() != WL_CONNECTED) {
    Serial.print(".");
    delay(300);
```

The OLED display on the device provides immediate, on-site status updates, while the Blynk app offers remote access to live data, pump status, and manual override options.

```
#include <WiFi.h>
#include <esp_now.h>


// Blynk credentials
#define BLYNK_TEMPLATE_ID "TMPL3g1Ucfp9p"
#define BLYNK_TEMPLATE_NAME "Control Centre"
#define BLYNK_AUTH_TOKEN "g6h4Sf-TNiMd5GAjvG2qF2cIBa8R9uDu"
#include <BlynkSimpleEsp32.h>
```

Once implemented, the farmer’s role in irrigation is significantly reduced—they will no longer need to manually assess soil or operate pumps.



Their remaining tasks include ensuring water availability in the source tank, maintaining the hardware components occasionally, checking the system status via the app, and intervening only if an alert indicates a fault or exceptional condition. This makes the system both autonomous and supportive, easing daily labour while keeping farmers in control.

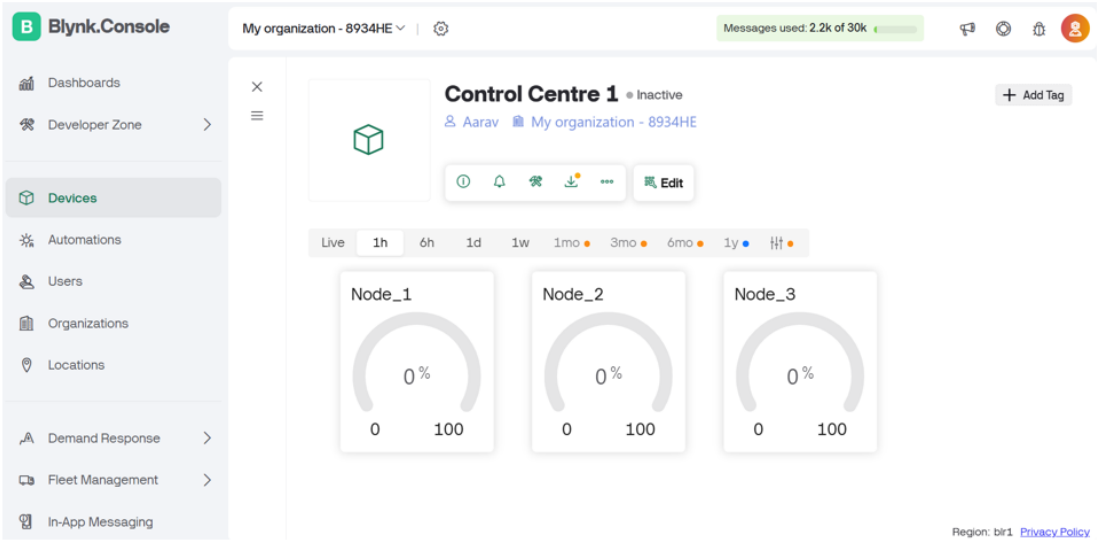
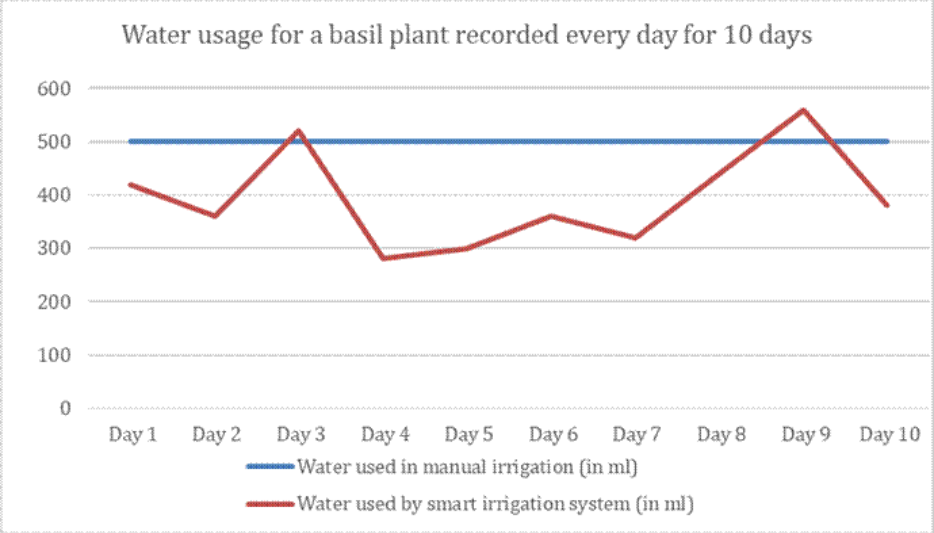


Fig: Blynk Dashboard

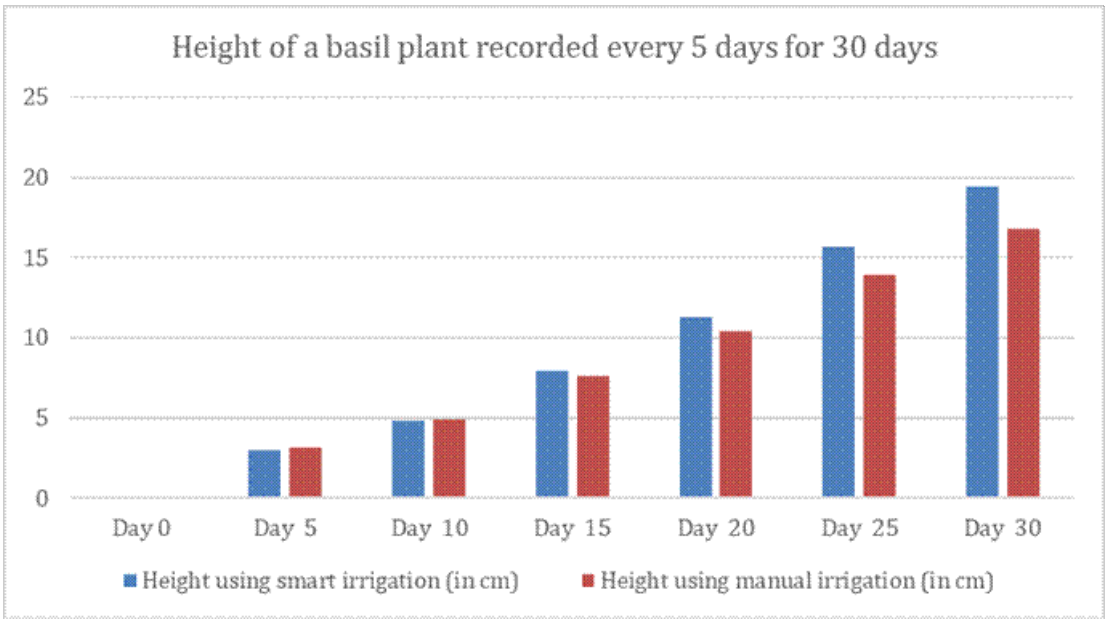
Results and Discussion

Water saving



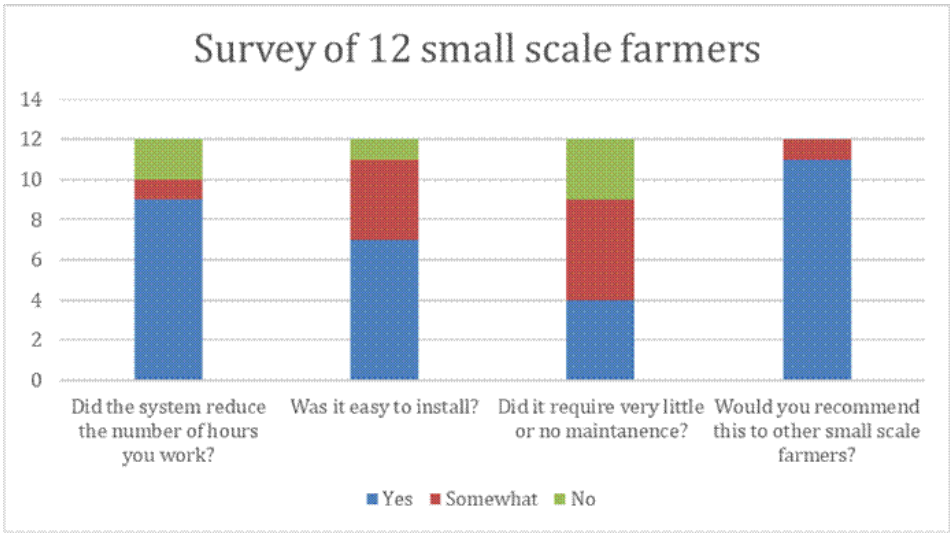
Manual irrigation used 5000ml of water while the smart irrigation system used 3940ml of water over 10 days. The smart irrigation system saved 1060 ml of water when set to keep soil moisture content above 40%. This reduced water usage by over 20% by optimizing irrigation and compensating varying quantities of water evaporation.

Plant growth (yield)



The height of the basil plant using manual irrigation was 16.8 cm while it was 19.4 using smart irrigation. The basil plant using the smart irrigation system grew 2.6cms taller. All other conditions were kept the same. This translates into a roughly 15% greater rate of plant growth and therefore significantly higher crop yield.

Survey



The 12 farmers originated from the outskirts of Delhi and were questioned in their native language (Hindi). They were told to select one option out of “yes”, “somewhat” and “no” and their answers were recorded. This graph shows a highly positive user response from farmers hence proving real-world feasibility and application of the product. The product, therefore, successfully improves quality of life for farmers by reducing working hours and is easy to install even with a rudimentary briefing of the usage. Farmers’ appreciation is also shown by the fact that 11 out of 12 farmers would recommend the product to others. However, the product did face slight maintenance problems and needed replacement of minor components.

Although water saving and plant growth tests were conducted on a basil plant in a pot, these findings can be translated to similar results for small-scale farmland. All the data revealed positive results and showed how the module successfully met its targets and satisfied all requirements.

Safety was prioritized throughout the design and implementation of this system. Low-voltage DC components were chosen to reduce electrical hazards, and protective 3D-printed casings were used to prevent damage and accidental contact with circuits. The water pump was tested under

controlled conditions to ensure it operated safely and reliably. From an ethical standpoint, the system was designed to be affordable, sustainable, and accessible to smallholder farmers with limited technical literacy. Surveys were conducted with informed consent and in the local language (Hindi), ensuring clear communication and respectful engagement with farmers. Data privacy and transparency were maintained during all phases of testing and feedback collection.

Conclusion

Objective	Extent to which achieved	How it was achieved/what could be improved
Improve quality of life for farmers by reducing hours spent on the field	Fully	Farmers reported reduced working hours in the survey. The device achieved its primary objective of aiding farmers in crop irrigation.
Reduce water wastage by adapting water quantity dispensed according to environmental conditions	Fully	Data collection revealed a 21% decline in water usage.
Improve crop yield by optimizing irrigation	Fully	Data collection revealed a 15% higher plant growth rate. Although, this number could vary and requires more extensive testing.

Create an easy to install module which can be operated with rudimentary knowledge of technology	Partially	The device was effectively applied but required an orientation session to explain its function. The farmers still had gaps in understanding which could be resolved through a simple manual.
Conduct a pilot study to compare crop yield, quality of life of farmers and water wastage with and without the module	Fully	Tests and studies were conducted to prove the success of the device by measuring 2 metrics as well as user feedback.
Enable farmers with modern technology	Partially	While the device served as an introduction to technologies like IoT and automation, this initial exposure alone is not sufficient for farmers to fully adopt and utilize modern technology effectively.

To enhance its utility and adoption, several future developments are proposed. Integrating automated fertilizer dispensing could convert the system into a fully autonomous crop care module. Adding a voice-based or local-language text alert system would improve accessibility for non-tech-savvy farmers. From a deployment standpoint, partnering with schools, NGOs, and agricultural outreach programs could expand the system's reach and encourage wider adoption in rural communities. Furthermore, creating a modular and open-source version could allow local innovators to adapt the system to regional needs and constraints.

References

1. **Arduino.cc Official Documentation:** This website helped me understand the basics of coding and allowed me to get started with programming. It included tips on how to integrate electrical components with code

2. **Tinkercad:** Simulate circuits before physically building them to not only help build understanding of electrical circuits but also reduced time consumption by allowing me to test circuits before building them
3. **Sparkfun and Adafruit Learning Resources:** I used these libraries while coding in Arduino.IDE
4. **YouTube: Paul McWhorter Arduino Series:** This playlist helped me garner a better understanding of electrical components
5. **"A Novel Autonomous Irrigation System for Smart Agriculture Using AI and 6G-IoT"**: This paper presents an autonomous irrigation system that leverages artificial intelligence (AI) and 6G-enabled Internet of Things (IoT) technologies. By analyzing historical weather data, the system predicts optimal irrigation schedules, aiming to enhance water efficiency and crop yield in smart agriculture applications. https://www.sciencedirect.com/science/article/abs/pii/S0141933123001497
6. **"Autonomous Irrigation System"**: This study details the development of an autonomous irrigation system utilizing multiple sensors, including soil moisture and humidity sensors, connected to a Raspberry Pi microcontroller. The system assesses real-time soil conditions to regulate water output, aiming to minimize water wastage and improve irrigation efficiency, particularly in regions with depleting water tables. https://www.ijrte.org/wp-content/uploads/papers/v7i6/F2797037619.pdf
7. **"The Effects of AI-Supported Autonomous Irrigation Systems on Water Efficiency and Plant Quality: A Case Study of *Geranium psilostemon* Ledeb"**: This research investigates the impact of an AI-supported irrigation system on the cultivation of *Geranium psilostemon* at Rize Recep Tayyip Erdoğan University. The AI system continuously monitors environmental parameters to optimize irrigation, resulting in improved water efficiency and plant growth compared to manual irrigation methods.
8. https://www.mdpi.com/2223-7747/14/5/770